JEE APEX FTS – JEE ADV. FULL SYLLABUS TEST-09 [P1]

KEY SHEET

MATHEMATICS

1	**A**	2	**A**	3	**C**	4	**B**	5	**BD**	6	**ABCD**
7	**ABC**	8	**101**	9	**0**	10	**8**	11	**100**	12	**5**
13	**6**	14	**A**	15	**B**	16	**C**	17	**A**		

PHYSICS

18	**A**	19	**C**	20	**B**	21	**B**	22	**ABC**	23	**AD**
24	**BCD**	25	**3**	26	**45**	27	**6**	28	**4**	29	**2**
30	**3**	31	**A**	32	**C**	33	**A**	34	**B**		

CHEMISTRY

35	**C**	36	**A**	37	**D**	38	**A**	39	**BC**	40	**ACD**
41	**ACD**	42	**5**	43	**8**	44	**7**	45	**4**	46	**3**
47	**7**	48	**C**	49	**A**	50	**A**	51	**B**		

SOLUTIONS
MATHEMATICS

1. $f(x)+f\left(\frac{1}{1-x}\right)=\frac{2}{x}-\frac{2}{1-x}$ .....(1)

Now Replace X By $\frac{1}{1-x}$ We Get

$f\left(\frac{1}{1-x}\right)+f\left(1=\frac{1}{x}\right)=2(1-x)-2\left(1-\frac{1}{x}\right)=-2x+\frac{2}{x}$ .....(2)

Replace x by $\frac{1}{1-x}$ we get

$f\left(1-\frac{1}{x}\right)+f(x)=\frac{2x}{x-1}-2x$ ....(3)

From equation (1) – (2), we get

Now, f(x) $-f\left(1-\frac{1}{x}\right)=2x-\frac{2}{1-x}$ ....(4)

From equation (3) + (4), we get

Again, 2f (x) = $\frac{2x}{x-1}-\frac{2}{1-x}$

$f(x)=\frac{x+1}{x-1}$ Which is a one-one function

2. Here h(x) =f(g(x))

And h'(x) = $f'(g(x))g'(x)<0\ \forall\ x\in[0,\infty)$

$g'(x)<0\ \forall x\in[0,\infty)$

$f'(g(x))>0\forall x\in[0,\infty)$

$Also\ h(0)=0$

Therefore, $h(x)<0\ \forall x\in[0,\infty)$

Now, p(x) = h$\left(x^3-2x^2+2x\right)-h(4)$

Or $p'(x)=h'\left(x^3-2x^2+2x\right)\left(3x^2-4x+2\right)<0\ \forall x\in(0,2)$

Since $h'\left(x^3-2x^2+2x\right)<0\forall x\in(0,\infty)$

and $3x^2-4x+2>0\ \forall x\in R$

$\Rightarrow$p(x) is a decreasing function

$p(2)<p(x)<p(0)$ for all $\in(0,2)$

$h(4)-h(4)<p(x)<h(0)-h(4)$

$\Rightarrow 0<p(x)<-h(4).$

3. $a^{10}b^{10}c^{10}c^{10}\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d}\right)^{10}$

$\therefore$ Required coefficient = coefficient of

$a^{-2}b^{-6}c^{-1}d^{-1}$ $in\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d}\right)^{10}=\frac{10!}{2!6!1!1!}=\frac{10\times9\times8\times7}{2}=2520$

4. Total ways = 4.5.6.7.8.9.10.11

$=\frac{11!}{3!}={}^{11}P_8$

5. $\lim\limits_{x\to0} f\left(x\right)=\lim\limits_{x\to0}\frac{P\left(\frac{1}{x^3}\right)}{\frac{1}{e^{x^4}}}=\lim\limits_{t\to\infty}\frac{P\left(t^3\right)}{e^{x^4}}=0$

As we know, $\lim\limits_{x\to0}\frac{x^n}{e^x}=0, n\in N)$

$\Rightarrow \lim\limits_{x\to0} f\left(x\right)=f\left(0\right)=0$

$f^{'}\left(0\right)=\lim\limits_{h\to0}\frac{P\left(\frac{1}{h^3}\right)}{\frac{1}{he^{h^4}}}$

$=\lim\limits_{h\to0}\frac{tP\left(t^3\right)}{e^{t^4}}=0$

6. $\left(px^2+qy^2+r\right)\left(4x^2+4y^2-8x-4\right)=0$

$\Rightarrow \quad 4x^2+4y^2-8x-4=0$

$\Rightarrow \quad \left(x-1\right)^2+y^2=1$

Or

$px^2+qy^2+r=0$ will represents

(a) Two straight lines if $r = 0$ and p,q are of opposite sign.

(b) a circle if $p = q$ and r is of opposite sign that of p.

(c) a hyperbola if p and q are of opposite sign and $r\neq0$.

(d) An ellipse if p and q are unequal but of same sign and r is of sign opposite to that of p.

7. $(a)\,\vec{r}=\left(\vec{r}.\hat{i}\right)\hat{i}+\left(\vec{r}.\hat{j}\right)\hat{j}+\left(\vec{r}\,.\hat{k}\right)\hat{k}$

Put $\vec{r}=\left(\vec{a}\times\vec{b}\right)$

$\Rightarrow$(a) is correct

(b) Put $\vec{r}=\vec{a}$ *and* $\vec{r}=\vec{b}$

Take $\vec{a}\,.\vec{b}\ \Rightarrow(b)$ is correct

(c) $\vec{u}=\hat{b}\times\left(\hat{a}\times\hat{b}\right)$

$|\vec{u}| = |\vec{a} \times \vec{b}| = |\vec{V}| \Rightarrow (c)$ is correct.

(d) is obviously wrong.

8. $x + \sin y = 101$

$x + 101\cos y = 100$

Equations $(i) - (ii)$

$\Rightarrow \sin y - 101\cos y = 1$

$\therefore \quad \sin y = 1 + 101\cos y$

$y \in \left[0, \frac{\pi}{2}\right]$

$\therefore \quad \cos y \in [0,1]$

$\therefore \quad$ *Possible only if* $\sin y = 1$

$\Rightarrow y = \frac{\pi}{2}$ *and* $\cos y = 0 \Rightarrow y = \frac{\pi}{2}$

$\therefore \quad \sin y = 1 \Rightarrow x = 100$

$\therefore \quad [x + y] = 101$

9. $S_n = \sum n(n+1) = \frac{1}{3}n(n+1)(n+2)$

$t_n = \frac{1}{n(n+1)(n+2)(n+3)}\left(\frac{(n+3)-n}{3}\right)$

$= \frac{1}{3}\left(\frac{1}{n(n+1)(n+2)} - \frac{1}{(n-1)(n+2)(n+3)}\right)$

$\sigma_n = \frac{1}{3}\left(\frac{1}{1}.\frac{1}{2}.\frac{1}{3} - \frac{1}{(n-1)(n+2)(n+3)}\right)$

$\sigma_{n-1} = \frac{1}{3}\left(\frac{1}{6} - \frac{1}{(n)(n+1)(n+2)}\right)$

$18S_n\sigma_{n-1}$

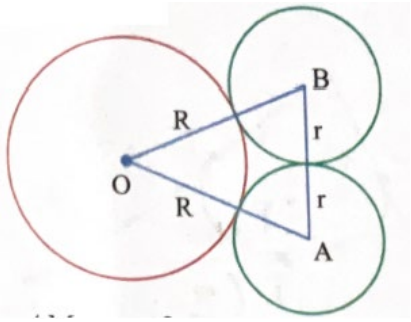
$= \frac{18}{3}.n(n+1)(n+2)\left(\frac{(n)(n+1)(n+2)-6}{18(n)(n+1)(n+2)}\right)$

$= \frac{(n)(n+1)(n+2)-6}{3}$

$18S_n\sigma_{n-1} - S_n = -2$

$18S_n\sigma_{n-1} - S_n + 2 = 0$

10. Let the small circles of centres A and B meet at



$$\frac{AM}{OA}=\sin\frac{2\pi}{24}=\sin 15^0$$

$$(R+r)\frac{\sqrt{3}-1}{2\sqrt{2}}=r,$$

$$\frac{R}{r}=\frac{2\sqrt{2}}{\sqrt{3}-1}\left(1-\frac{\sqrt{3}-1}{2\sqrt{2}}\right)$$

$$=\frac{\left(\sqrt{3}-1\right)}{2}\left(2\sqrt{2}-\sqrt{3}+1\right)=\sqrt{6}+\sqrt{2}-1$$

$$\Rightarrow m=6,n=2,m+n=8$$

11. Let $S=1+\frac{2}{z}+\frac{3}{z^2}+\frac{4}{z^3}+.....$

$$\frac{S}{z}=+\frac{1}{z}+\frac{2}{z^2}+\frac{3}{z^3}+.....$$

$$S\left(1-\frac{1}{z}\right)=1+\frac{1}{z}+\frac{1}{z^2}+\frac{1}{z^3}.....$$

$$\frac{S(z-1)}{z}=\frac{1}{1-\frac{1}{z}}=\frac{z}{z-1}$$

$\therefore \quad iz^2=\frac{z^2}{(z-1)^2} \quad (z\neq 0) \qquad i=\frac{1}{(z-1)^2}$

$\Rightarrow \quad (z-1)^2=\frac{1}{i}=-i \qquad \Rightarrow \quad z-1=\pm\sqrt{i}$

$$z=1\pm\sqrt{-i}$$

Given, $z=n\pm\sqrt{-i} \qquad \Rightarrow \quad 100n = 100$

12. $\sum_{k=1}^{n}a_{ki}.a_{kj}=0$

$$i=1\sum_{k=1}^{n}a_{ki}=0\left(as\ a_{ki}=1\ \forall k\right)$$

So in j[th] column sum of elements must be zero. So there must be equal numbers of +1 and -1 say 'm' so $n = 2m$

Putting $j=1\sum_{k=1}^{n}a_{ki}=0$ so similarly there exist one more such i^{th} column with m '+1' and m '-1' elements.

Now for the above i^{th} and j^{th} column $\sum a_{ki}.a_{kj}$ will be have terms of the form

$1\times 1, 1\times(-1), (-1)\times 1 \; and \; (-1)\times(-1)$

If we assume r and $m-r$ no. of i s come in last two forms. So

$r-(m-r)-(m-r)+r=0$

$\Rightarrow m=2r$

$So \; n=4r$

Hence n must be multiple of 4. So $m = 25$

13. Let $P(E_1)=x, P(E_2)=y, P(E_3)=z$

then $x(1-y)(1-z)=p$

$x(1-y)(1-z)=\alpha$

$(1-x)y(1-z)=\beta$

$(1-x)(1-y)z=\gamma$;

so $\frac{1-x}{x}=\frac{p}{\alpha}\Rightarrow x=\frac{\alpha}{\alpha+p}$

similarly $z=\frac{\gamma}{\gamma+p}$ $\qquad so \quad \frac{P(E_1)}{P(E_3)}=\frac{\frac{\alpha}{\alpha+p}}{\frac{\gamma}{\gamma+p}}=\frac{\frac{\gamma+p}{\gamma}}{\frac{\alpha+p}{\alpha}}=\frac{1+\frac{p}{\gamma}}{1+\frac{p}{\alpha}}$

Also given

$\frac{\alpha\beta}{\alpha-2\beta}=p=\frac{2\beta\gamma}{\beta-3\gamma}\Rightarrow\beta=\frac{5\alpha\gamma}{\alpha+4\gamma}$

Substituting back

$\left(\alpha-2\left(\frac{5\alpha\gamma}{\alpha+4\gamma}\right)\right)p=\frac{\alpha.5\alpha\gamma}{\alpha+4\gamma}$

$\Rightarrow \alpha p-6p\gamma=5\alpha\gamma$

$\Rightarrow\left(\frac{p}{\gamma}+1\right)=6\left(\frac{p}{\alpha}+1\right)\Rightarrow\frac{\frac{p}{\gamma}+1}{\frac{p}{\alpha}+1}=6$

14. A) When $-1\le x<0$.

$f(x)\in[0,1) \; and \; f(x)=x+1$

$\therefore \; f(f(x))=(x+1)^2-1=x^2+2x$

B) $\frac{(\cos 2x+1)(\sec^2 x+2\tan x)}{2}$

$$=\left(\frac{1-\tan^2 x}{1+\tan^2 x}+1\right)\left(\frac{1+\tan^2 x+2\tan x}{2}\right)$$

$$=\frac{1+\tan^2 x+2\tan x}{1+\tan^2 x}$$

$$i.e.\ f\left(\frac{2\tan x}{1+\tan^2 x}\right)=1+\frac{2\tan x}{1+\tan^2 x}$$

$$\therefore\ \ f(x)=1+x$$

$$x=0, y=0, f(1)=(1+1)^2=2^2$$

$$x=0, y=1, f(2)=(1+2)^2=3^3$$

$$f(x)=(x+1)^2$$

$$4<x<5, \left[\frac{x}{4}\right]=1$$

$$\Rightarrow f(x)=2x+3$$

$$\therefore\ y=2x+3\Rightarrow x=\frac{y-3}{2}$$

$$\therefore f^{-1}(x)=\frac{x-3}{2}$$

15. $A^TD=DA^T$

$DD^T=D^TD=I$

$\Rightarrow \quad DD^TA^TD=DA^T \qquad DA^{-1}=DA^T\Rightarrow A^{-1}=A^T$

$\Rightarrow \qquad\qquad BC^2=B^T \qquad B=B^T$

(A) $B=B^{-1} \qquad \Rightarrow\ B^2=I$

(B) $\left(ACA^T\right)^{2019}=\left(ACA^T\right)\left(ACA^T\right).....\left(ACA^T\right)=AC^{2019}A^T=AC^kA^T \qquad \Rightarrow k$ is odd

(C) $\left|ACA^T\right|=|A|^2|C|=|C|=\pm 1$

(D) $\left|BB^T\left(B^{-1}\right)^2\right|+\left|\left(AA^T\right)^{2018}\right|=1+1=2$

16. A) $x^2(ydx+xdy)=\dfrac{xdy-ydx}{y^2}=(xy)\left(\dfrac{x}{y}\right)d(xy)=d\left(\dfrac{x}{y}\right)\dfrac{x^2y^2}{2}=1h\left(\dfrac{x}{y}\right)+k$

B) $2x^3ydy+\left(1-y^2\right)\left(x^2y^2+y^2-1\right)dx=0$

$$\frac{2y}{\left(1-y^2\right)^2}=\frac{dy}{dx}+\frac{y^2}{1-y^2}\frac{1}{x}=\frac{1}{x^3} \qquad Put\ \ \frac{y^2}{1-y^2}=4 \quad \frac{dy}{dx}+\frac{y}{x}=\frac{1}{x^3}$$

C) Apply C & D $\dfrac{dy}{dx}=\dfrac{e^{-x}}{e^x}=e^{-2x}$

D) $In\dfrac{dy}{dx}=3x+4y\Rightarrow\dfrac{dy}{dx}=e^{3x+4y}$

17. Conceptual Based

18. $V = \frac{s}{t}$

$$\Delta v = \frac{1}{t}.\Delta s + s\frac{\Delta t}{t^2}$$

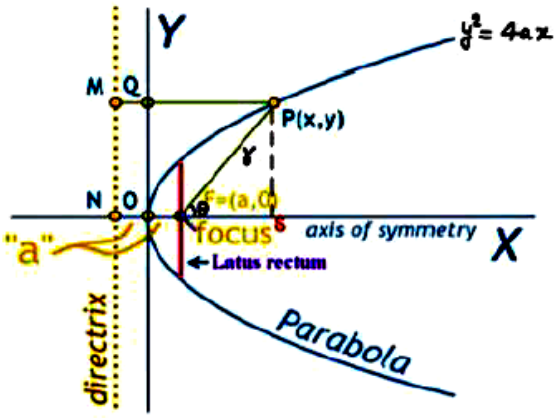
$$= \frac{\Delta s}{t}.\frac{\Delta t}{\Delta t} + S\frac{\Delta s}{\Delta s}\frac{\Delta t}{\Delta t}\frac{\Delta t}{t^2}$$

$$= .\frac{\Delta s}{\Delta t}\left(\frac{\Delta t}{t} + \frac{s\Delta t^2}{\Delta s.t^2}\right)$$

$$= \Delta v = \pm\frac{\Delta s}{\Delta t}$$

$$\left(\frac{\Delta t}{t} + \frac{s\Delta t^2}{\Delta s.t^2}\right) = \pm 1$$

19. Geometry of Parabola



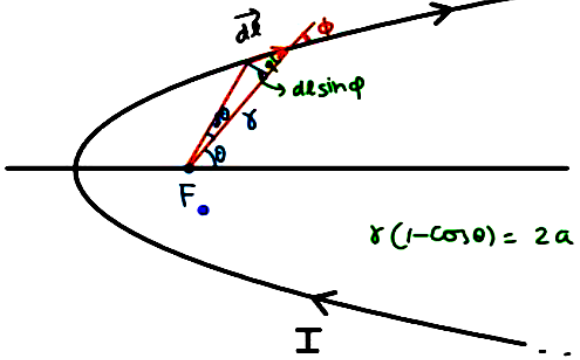
γ = distance between point on parabola and focus.

PF = PM

$\gamma = \gamma\cos\theta + 2a$

$\boxed{\gamma(1-\cos\theta) = 2a}$

Parametric eqn of Parabola



$$d\vec{B} = \frac{\mu_0 i}{4\pi}\frac{d\vec{l}\times\vec{\gamma}}{\gamma^3}$$

$$dB = \frac{\mu_0 i}{4\pi}\frac{(dl)(\gamma)\sin\phi}{\gamma^3}$$

$$= \frac{\mu_0 i}{4\pi}\frac{(dl)\sin\phi}{\gamma^2}$$

$$= \frac{\mu_0 i}{4\pi}\frac{\gamma\, d\theta}{\gamma^2}$$

$$dB = \frac{\mu_0 i}{4\pi}\frac{d\theta}{\gamma}$$

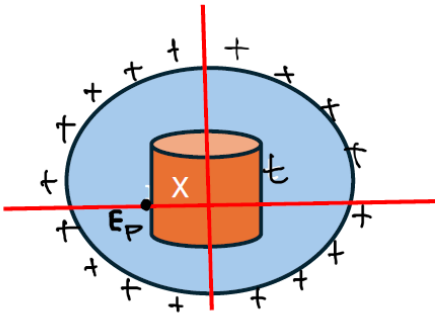
$$B = \int_0^{2\pi}\frac{\mu_0 i}{4\pi}\frac{d\theta}{2a}(1-\cos\theta)$$

$$B = \frac{\mu_0 i}{8\pi a}(\theta - \sin\theta)_0^{2\pi}$$

$$= \frac{\mu_0 i}{8\pi a}2\pi = \frac{\mu_0 i}{4a}$$

20. Calculation of field in the plane of the ring :
here the x<< Ro.
Imagine a hypothetical cylindrical Gaussian surface of radius x and length 2t and placed at the centre of Ring and perpendicular to the plane of the ring . Axis of the cylinder is parallel to the axis of ring

Electric field at a distance t on the axis of ring is $E = \dfrac{kQ_0 t}{\left(R^2 + t^2\right)^{\frac{3}{2}}}$



Apply Gauss law : $2\phi_1 + \phi_c = 0$ because on charge at centre of ring

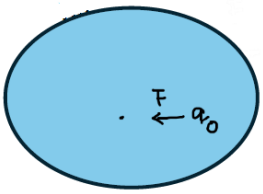
$2E\pi x^2 + E_p 2\pi x.2t = 0$

$E_p = \dfrac{-Ex}{2t}$

$= -\dfrac{kQ_0 t}{\left(R^2 + t^2\right)^{\frac{3}{2}}} \cdot \dfrac{x}{2t}$ R is very large as compared to t

$E_p = \dfrac{kQ_0 x}{2R^3}$ Direction towards the centre of ring

Analysing the motion of the charge particle in the plane of the ring:



1) Force is restoring in nature
2) Motion of both ring and point charge will be SHM .
3) $F = q_0 E_p \Rightarrow F = \left(\dfrac{kQ_0 q_0}{2R^3}\right).x$
4) $F = K_0 x$

After comparison $K_0 = \dfrac{kQ_0 q_0}{2R^3}$

Calculation of time period and analogy with a spring connected to two masses:



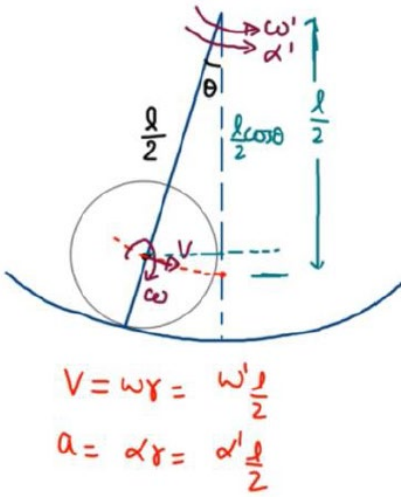
Time period of the system $T = 2\pi\sqrt{\dfrac{M_1 M_2}{\left(M_1 + M_2\right)k_0}}$

$T = 2\pi\sqrt{\dfrac{M_0 m_0 .2R_0^3}{\left(M_0 + m_0\right)kQ_0 q_0}}$

Time after which particle return to the centre

$$T=\frac{T}{2}=\pi\sqrt{\frac{M_0m_0.2R_0^3}{(M_0+m_0)kQ_0q_0}}\quad\text{Here}\quad K=\frac{1}{4\pi\in_0}$$

21. Energy Conservation



$$KE_{disc}=\frac{1}{2}mv^2+\frac{1}{2}\frac{mr^2}{2}\frac{v^2}{r^2}=\frac{3}{4}mv^2=\frac{3}{4}m\left(\frac{\omega^1 l}{2}\right)^2$$

$$PE_{disc}=mg\frac{l}{2}(1-\cos\theta)$$

$$mg\frac{l}{2}(1-\cos\theta)+\frac{3}{4}m\frac{l^2}{4}\omega^{1^2}+\frac{1}{2}k\left(\frac{l}{2}\theta\right)^2=\text{co}nst$$

$$\frac{mgl}{2}\sin\theta\not{\omega}^1+\frac{3}{4}m\frac{l^2}{4}2\not{\omega}^1\alpha^1+\frac{1}{2}k\frac{l^2}{4}2\theta\not{\omega}^1=0$$

$$\downarrow$$

$$\theta$$

$$\alpha^1=-\left(\frac{2k}{3m}+\frac{4g}{3l}\right)\theta$$

$$\omega^2=\frac{2k}{3m}+\frac{4g}{3l}$$

22 $P=VI$

$$P=50\times10^3\times20\times10^{-3}=1000w$$

Power converted into heat =990w

$$ms\Delta T=\Delta Q$$

$$ms\frac{\Delta T}{\Delta t}=\frac{\Delta Q}{\Delta t}$$

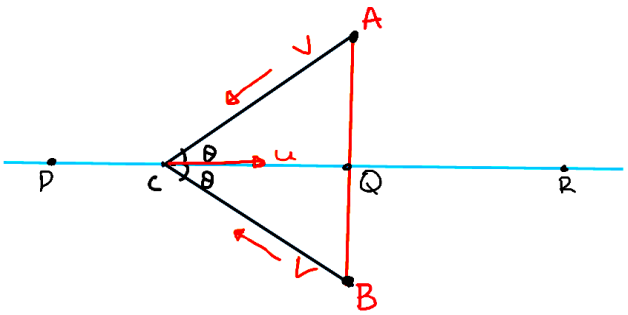
$$\frac{\Delta T}{\Delta t}=\frac{990}{495}=2^0C/\sec$$

Now $\frac{hc}{\lambda_m}=eV$

$$\lambda_{\min imum}=\frac{hc}{eV}$$

$$\lambda_{\min imum}=0.25\times10^{-10}m$$

23.



PQ=QR =1800m

Let C denote the portion of the car at any instance, and let θ denote angle shown in the figure. At this position. The frequencies heard from the sources A and B are respectively .

$f_A = \left(\frac{v + u\cos\theta}{v}\right) 118\,Hz$ and

$f_A = \left(\frac{v + u\cos\theta}{v}\right) 121\,Hz$

And the beat frequency is

$f_{beat} = f_B - f_A = \left(\frac{v + u\cos\theta}{v}\right) 3\,Hz$

At $P,\ \theta \approx 0$. Thus the beat frequency at P is

$f_P = \left(\frac{v+u}{v}\right) 3Hz = \frac{104}{99} \times 3Hz$

At $Q,\ \theta \approx \frac{\pi}{2}$. Thus the beat frequency at Q is

$f_Q = 3Hz$

At $R,\ \theta \approx \pi$. Thus the beat frequency at R is

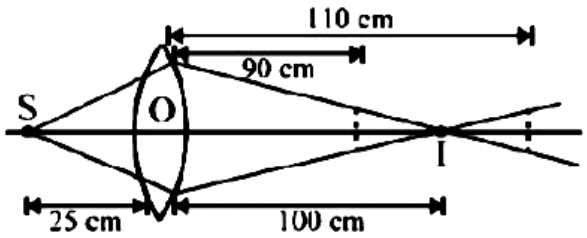
$f_R = \left(\frac{v-u}{v}\right) 3HZ = \frac{94}{99} \times 3Hz$

The value of θ changes most rapidly as θ approaches 90^0 , hence A is correct

$f_P + f_R = 6Hz = 2f_Q \Rightarrow C$ is incorrect

$f_P - f_R = \left(\frac{10}{99}\right) 3Hz = \frac{1}{10} f_Q \Rightarrow D$ is correct

24. $\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} - \frac{1}{(-25)} = \frac{1}{20} \Rightarrow \frac{1}{v} = \frac{1}{20} - \frac{1}{25} = \frac{1}{100} \Rightarrow v = 100\,cm$



From O to I intensity increases and then decreases. At x=90 cm and 110 cm intensity is same . (D) Radius at x=200 cm is equal to radius of lens .

25. Rate of collisions with vessel wall is

$N_c = \frac{1}{6} \times$ number density $\times\ V_{rms}$

$$N_c = \frac{1}{6} \times \frac{N}{V} \times v_{rms}$$

$$N_c = \frac{1}{6} \times \frac{N}{V} \times \sqrt{\frac{3RT}{M}}$$

As the rate of collisions with vessel wall is constant

$$N_c \alpha \frac{\sqrt{T}}{V} = const$$

$$\frac{T}{V^2} = \text{co}\, nst \ \ldots\ldots\ldots\ldots 1$$

As $PV = nRT \Rightarrow \frac{T}{PV} = \cos nt$ ……..2

From equations 1 and 2, $PV^{-1} = \text{co}\, nst$

Molar heat capacity is given as

$$C = \frac{R}{\gamma - 1} + \frac{R}{1-n}$$

$$C = C_v + \frac{R}{1-n}$$

$$C = \frac{5R}{2} + \frac{R}{1-(-1)}$$

C=3R

$\therefore x = 3$

26.



By force balance in y direction

N sin θ = mg

By force balance in x direction

N cos θ = R

By the torque balance about the point at hinge:

$$mg.\frac{l}{2}\sin\theta = Nr\cot\theta$$

After solving

$$\frac{Cos\theta}{\sin^3\theta} = \frac{l}{2r} = \frac{40}{2(10)}$$

$$\cot\theta = 2.Sin^2\theta = \frac{2}{\csc^2\theta} = \frac{2}{1+\cot^2\theta}$$

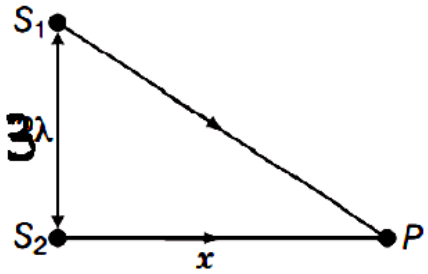
$$\cot^3\theta + \cot\theta = 2$$

$$(\cot\theta - 1)(\cot^2\theta + \cot\theta + 2) = 0$$

$$\cot\theta = 1$$

$$\theta = 45$$

27.



The path difference for minima are

$\frac{\lambda}{2},\frac{3\lambda}{2},\frac{5\lambda}{2},.....$

The maximum path difference is 3λ. Hence the minimum position obtained at $\frac{5\lambda}{2}$

$S_1P - S_2P = \frac{5\lambda}{2}$

$\sqrt{9\lambda^2 + x^2} - x = \frac{5\lambda}{2}$

after solving

$x = \frac{11\lambda}{20}$

$\frac{11\lambda}{20} = \frac{(n+5)\lambda}{20}$

$n = 6$

28. $R = \frac{u^2 \sin 2\theta}{g} = 10$

$u = 10$

$M_p = \frac{4}{3}\pi R_p^3 \times 2\rho$

$M_p = \frac{4}{3}\pi R_p^3 \frac{2M_e}{\frac{4}{3}\pi R_e^3}$

$M_p = \frac{2M_e R_p^3}{R_e^3}$

$V_p = \sqrt{\frac{2GM_p}{R_p}}$

$V_p = \sqrt{\frac{2G \times 2M_e R_p^3}{R_p \times R_e^3}}$

$R_p = 4 \times 10^3 m = 4Km$

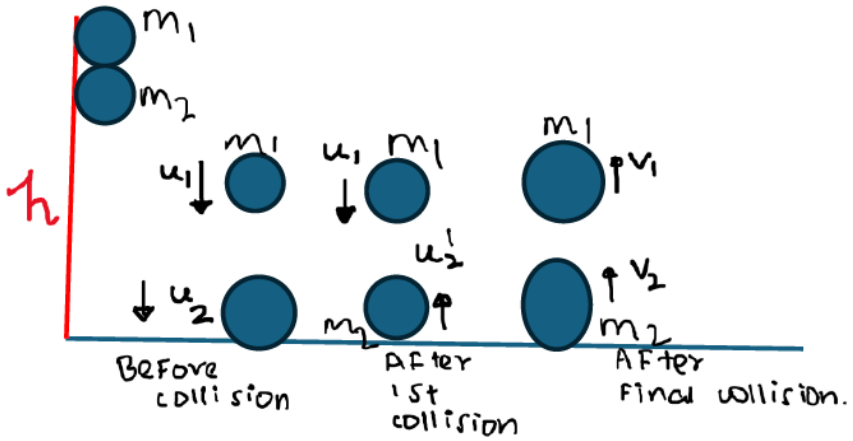
29. Torque of hydrostatic force about centre of sphere is already zero as hydrostatic force passes through the Centre. For translational equilibrium, the horizontal trust provided by both liquid should be equal and opposite.

$F_{LHS} = F_{RHS}$

$$\frac{1}{2}(2\rho)(g)h^2 = \frac{1}{2}(3\rho)(g)R^2$$

$$h = \sqrt{\frac{3}{2}}R = \sqrt{\frac{3}{2}}\sqrt{\frac{8}{3}} = 2m$$

30. Let m_1 = km and m_2 = m $\quad \therefore \frac{m_1}{m_2} = k$



$u_1 = u_2 = u = \sqrt{2gh}$

Collision is elastic hence velocity of m2 after collision with ground also be same

$u_2' = u$

Apply conservation of linear momentum

$m_1(-u) + m_2 u = m_1 v_1 + m_2 v_2$

$Km(-u) + mu = kmv_1 + mv_2$

$u(1-k) = v_2 + kv_1 \longrightarrow 1$

$$e = 1 = \frac{v_1 - v_2}{2u} \Rightarrow v_1 - v_2 = 2u$$

$v_2 = v_1 - 2u \longrightarrow 2$

Sub equation 2 in equation 1

$u(1-k) = v_1 - 2u + kv_1$

$$v_1 = \frac{u(3-k)}{k+1} \longrightarrow 3$$

Initially, total energy

$$E = \frac{1}{2}km\,u^2 + \frac{1}{2}mu^2$$

$$E = \frac{1}{2}m(2gh)(k+1) \longrightarrow 4$$

Final kinetic energy of upper ball

$$E_1 = \frac{1}{2}kmu^2\left(\frac{3-k}{K+1}\right)^2$$

$$E_1 = \frac{1}{2}km(2gh)\left(\frac{3-k}{K+1}\right)^2$$

$$E_1 = mghk\left(\frac{3-k}{k+1}\right)^2 \longrightarrow 5$$

Fraction of energy

$$\frac{E_1}{E} = \frac{K(3-k)^2}{(k+1)^3}$$

For maximum

$$f(k) = \frac{K(3-k)^2}{(k+1)^3} = \frac{9k+k^3-6k^2}{(k+1)^3}$$

$$f'(k) = 0$$

$$\frac{\partial}{\partial k}\left(\frac{9k+k^3-6k^2}{(k+1)^3}\right) = 0$$

After solving the above equation

$K = 1/3$ *and* 3

For max k = 1/3

$$\therefore k = \frac{m_1}{m_2} = \frac{1}{3}$$

31. $(a) \to s$

$(b) \to s$

$(c) \to r$

$(d) \to s$

(a)

$W_{ab} = area\ under\ p-V\ graph$

$$= \frac{3p_OV_O}{2} = 1.5p_OV_O$$

(b)

$$\Delta U_{ab} = Q_{ab} - W_{ab}$$

$$= 6p_OV_O - \frac{3p_OV_O}{2} = 4.5P_OV_O$$

(C)

$$C = \frac{Q}{n\Delta T} = \frac{W+\Delta U}{n\Delta T}$$

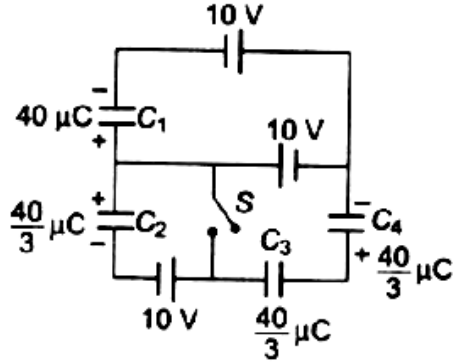
$$= \frac{6p_OV_O}{n\left(\frac{p_bV_b}{nR} - \frac{p_aV_a}{nR}\right)} = 2R$$

(D)

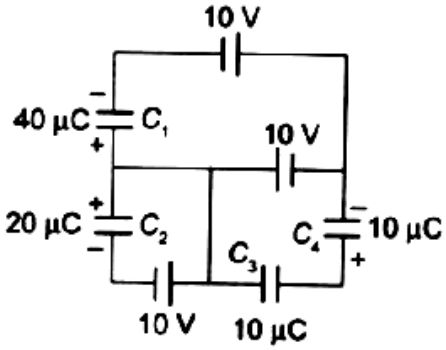
$$C_V = \frac{\Delta U}{n\Delta T} = \frac{4.5p_OV_O}{n\left(\frac{p_bV_b}{nR} - \frac{p_aV_a}{nR}\right)} = 1.5R$$

32.

Before closing the switch:



After closing the switch:



So charge on C_1 remains same, on C_2 increases and an C_3 and C_4 decreases.

33. (a) and (b)

$$v = \frac{\mu_2}{\mu_1}u$$

$$\Rightarrow |v| < |u|$$

$as \qquad \mu_1 > \mu_2$

i.e v *and* u are of same sign.

Or they are on same side of plane surface.

From plane surface, if object is real, image is virtual and vice-versa.

(c) and (d)

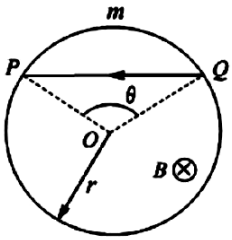
$$v = \frac{\mu_1}{\mu_2}u$$

$$\Rightarrow \qquad |v| > |u|$$

Other explanations are same.

34. Area $OPMQO = \frac{1}{2}r^2\theta$ Flux in this area, $\phi 1 = \frac{1}{2}r^2\theta B$

Induced emf in this area, $e_1 = \frac{d\phi_1}{dt} = \frac{1}{2}r^2\theta\frac{dB}{dt}$



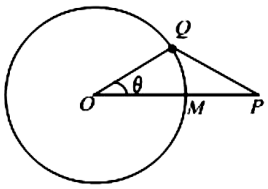
Area OPQ = r sin $\left(\frac{\theta}{2}\right) r\cos\left(\frac{\theta}{2}\right) = \frac{1}{2}r^2\sin\theta$

Induced emf in OPQ, $e_2 = \frac{d\phi_2}{dt} = \frac{1}{2}r^2\sin\theta\frac{dB}{dt}$

e_2 will be only in part PQ, because in ***OQ*** and ***OP*** induced emf will be zero.
Clearly, $e_2 < e_1$, because are OPQ < area $OPMQO$
Since B is increasing, so e.m.f will be in anticlockwise direction. Hence and P will be positive with respect to Q

1. Here emf in OPQ will be due to flux changing in area OMQ. This area $\frac{1}{2}r^2\theta$.

The entire emf will be in part PQ. End Q will be positive.

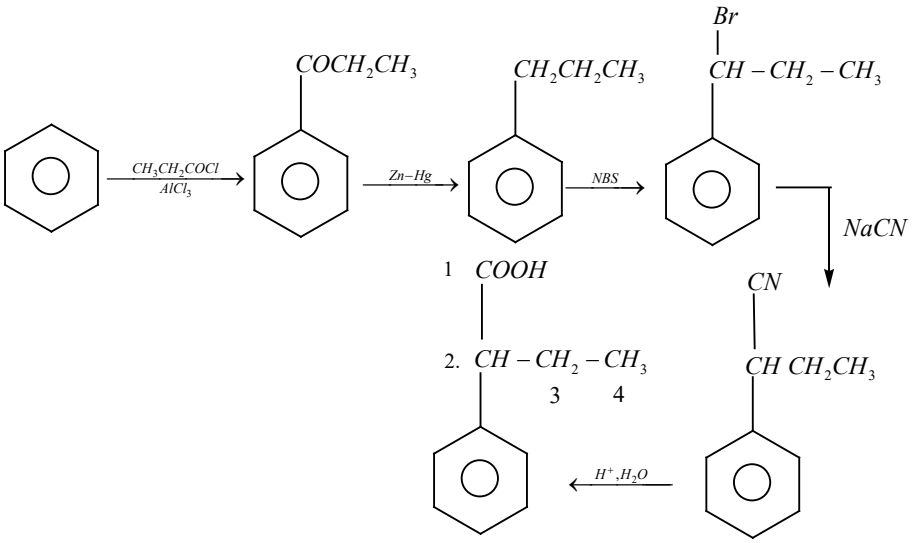


1. Induced emf = $\frac{1}{2}r^2\theta\frac{dB}{dt}$. End P will be positive

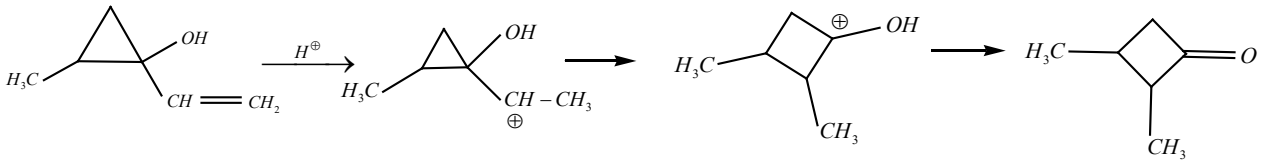
Area in which flux is changing is less than $\frac{1}{2}r^2\theta$ End Q will be positive.

CHEMISTRY

35.



36.



37. $5SO_3^{-2} + 2MnO_4 + 6H^+ \rightarrow 2Mn^{+2} + 5SO_4^{-2} + 3H_2O$

$2MnO_4^- + 5NO_2^- + 6H^+ \rightarrow 2Mn^{+2} + 5NO_3^- + 3H_2O$

$2MnO_4^- + H_2S + 6H^+ \rightarrow 2Mn^{+2} + 5S + 8H_2O$

38. Let total no. of moles taken = n

No. of moles in condensate = $\frac{3}{4}n.$

No. of moles in residue $= \frac{1}{4}n.$

No. of moles of A in the original mixture = No of moles of A in the condensate + No of moles of A in residue

$$n_A = \frac{3}{4}n \times 0.75 + \frac{1}{4}n \times 0.3$$

$$\frac{n_A}{n} = \frac{2.55}{4} = 0.6375$$

39. Phenol is more reactive in basic medium towards coupling reaction because it exists as phenoxide ion in basic medium. Aniline prefer coupling in dilute acidic medium

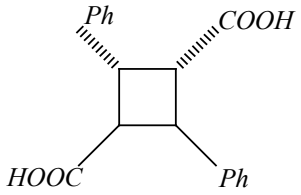
40. Conceptual

41. Y_2 must be either $Cl_2 or\ F_2$

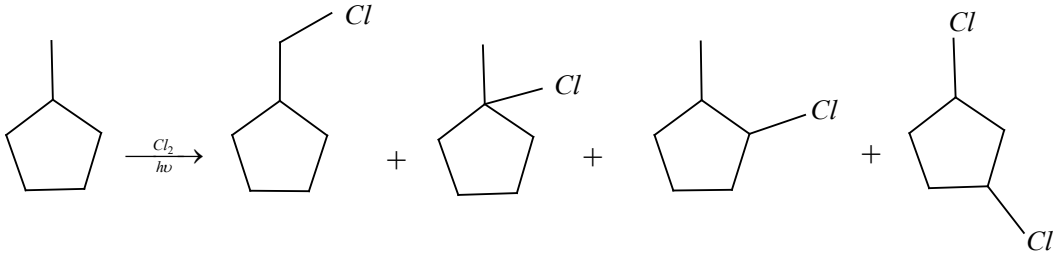
X_2 must be iodine

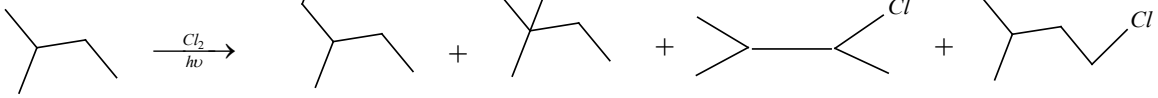
S.R. P $F_2 > Cl_2 > Br_2 > I_2$

42.



43.

X = 

Y = 

$X + Y$

$4 + 4 = 8$

44. For paramagnetic compounds weight increases in the applied magnetic field

$NO, NO_2, O_2, K_4\left[Fe(CN)_6\right], KO_2, MnSO_4, NiSO_4, CuSO_4$ are paramagnetic in nature

45. V_{O_2} in the initial mixture = V

$V_{H_2} = 0.03 - V$ $\qquad 2H_2 + O_2 \rightarrow 2H_2O$

$\left[0.03 - V\right]$ lits of H_2 can consume $\dfrac{0.03 - V}{2}$ lit of O_2

Volume of O_2 left over $= V - \left(\dfrac{0.03 - V}{2}\right) = 0.003$

Solving $v = 0.012$ $\qquad$ % V_{O_2} in the mixture

$= \dfrac{0.012}{0.03} \times 100 = 40$

46. In rock salt like structure

$2\left(r_c + r_a\right) = a \qquad r_c + r_a = \dfrac{a}{2}$

Since $\dfrac{r_c}{r_a} = \dfrac{1}{2} \qquad r_c = \dfrac{r_a}{2} \qquad \dfrac{r_a}{2} + r_a = \dfrac{a}{2} \qquad \dfrac{3r_a}{2} = \dfrac{a}{2}$

$r_a = \dfrac{a}{3} \qquad r_c = \dfrac{a}{6} \qquad P.F = 4 \times \dfrac{4}{3} \dfrac{\pi\left(r_c^3 + r_a\right)^3}{a^3} = 0.7$

Percentage of void = 0.3

47. When volume occupied by molecule is negligible $v - b \simeq v$

$\left(P + \dfrac{a}{v^2}\right) v = RT$

$P = \dfrac{RT^2}{4a} = 34.88$ atn

$34.88 = 5x$

$\Rightarrow x = 7$

48. 2.4 - θNP Test : Aldehydes & Ketones

Tollens Test : Aldehyde, Formic Acid

49. A = a = q, st, b = r, c = p , d =r
Conceptual

50. Conceptual

51. $\alpha = \frac{i-1}{n-1}$ $R.L.V.P = \frac{in_1}{in_1 + n_2}$ $\pi = icRT$ $\Delta_{T_f} = i \times k_f \times m$